Bureau International des Poids et Mesures

Le système international d'unités
The International System of Units

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Annexe 2
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Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Strontium 88 Ion ()

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CCL-CCTF Frequency Standards Working Group

April 2018
avril 2018

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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
Part 2: Strontium 88 Ion ()

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1. Recommended value [1] of the frequency in the CIPM List of Frequencies

$f\left({}^{88}\mathrm{S}r^{+}\right) = 444\ 779\ 044\ 095\ 486.5\ \mathrm{Hz}$

equivalent to

$\lambda\left({}^{88}\mathrm{S}r^{+}\right) = 674\ 025\ 590.863\ 134\ \mathrm{fm}$,

with a relative standard uncertainty of 1.5×10^{-15}.

This radiation was endorsed by the CIPM as a secondary representation of the definition of the second [2].

2. Source data

Adopted value $f\left({}^{88}\mathrm{Sr}^{+}\right) = 444\ 779\ 044\ 095\ 486.5$ Hz

With relative standard uncertainty: 1.5×10^{-15}

calculated from

$f\left({}^{88}\mathrm{Sr}^{+}\right)$/ Hz	u/ Hz	source data
444 779 044 095 484.6	1.5	[3]
444 779 044 095 484	15	[4]
444 779 044 095 485.5	0.9	[5]
444 779 044 095 486.71	0.24	[6] [(a)]
444 779 044 095 485.27	0.75	[7]

(*a*) Since the result is largely dominated by the single measurement of ref. [6], the final uncertainty has been increased by a factor of three following the procedures outlined in ref. [8].

as a weighted mean of the above values.

$^{88}Sr^{+}$, 5s $^{2}S_{1/2}-$4d $^{2}D_{5/2}$ unperturbed optical transition

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